Alexei Oblomkov

Professor of Mathematics

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Research Interests

Representation theory (double affine Hecke and symplectic reflection algebras, Cherednik algebras); algebraic geometry (Hilbert schemes, Gromov–Witten and Donaldson–Thomas invariants, affine Springer fibers); knot homology and categorification; mathematical physics and integrable systems.

Education

2005	Ph.D. in Mathematics, Massachusetts Institute of Technology. Advisor: Pavel Etingof.
1998	M.A. in Pure Mathematics, Independent University of Moscow and Moscow State University.

Employment

2020–	Professor, University of Massachusetts, Amherst.
2018 (Spring)	Research Professor, MSRI, Berkeley.
2015–2019	Associate Professor, University of Massachusetts, Amherst.
2009–2015	Assistant Professor, University of Massachusetts, Amherst.
2006–2009	Instructor, then Visiting Assistant Professor, Princeton University.
2005–2006	Postdoctoral Fellow, Institute for Advanced Study.

Grants and Awards

2022–2025	NSF DMS-2200798, $217,038, sole PI.
2018–2021	NSF FRG DMS-1760373, $300,000 for UMass.
2014–2019	NSF CAREER DMS-1352398, $420,000, sole PI.
2010–2013	NSF DMS-1001609, $123,910, sole PI.
2007–2010	NSF DMS-0701367, $111,300, sole PI.
2018	Simons Fellow in Mathematics.
2014	NSF CAREER Award.
2010–2012	Alfred P. Sloan Research Fellowship.

Selected Publications

Selected Preprints and submitted work

6. *Asymptotic q,t-Fuss–Catalan numbers for type B*, arXiv:2609.11691.
5. (with Y. Huang, R. Jiang) *Quot scheme of points on torus knot singularities*, arXiv:2608.16086.
4. (with I. Cavey, E. Gorsky, J. Turner) *Wall-crossing for Hilbert schemes*, arXiv:2607.05624.
3. (with L. Rozansky) *A categorification of representations of $U_q(\mathfrak{gl}_{1|1})$*, arXiv:2510.13039.
2. (with I. Cavey, E. Gorsky, J. Turner) *Newton–Okounkov bodies for nested Hilbert schemes*, arXiv:2510.07420.

1. (with O. Kivinen, G. Wyss) *Discriminants and motivic integration*, arXiv:2503.18449.

Selected Published and accepted papers

8. (with E. Gorsky, O. Kivinen) *The affine Springer fiber – sheaf correspondence*, Advances in Mathematics **464** (2025), arXiv:2204.00303.
7. (with L. Rozansky) *Categorical Chern character and braid groups*, Adv. Math. **437** (2024), 66 pp., arXiv:1811.03257.
6. (with M. Moreira, A. Okounkov, R. Pandharipande) *Virasoro constraints for stable pairs on toric threefolds*, Forum Math. Pi **10** (2022), Paper No. e20, 62 pp., arXiv:2008.12514.
5. (with L. Rozansky) *Knot homology and sheaves on the Hilbert scheme of points on the plane*, Selecta Math. (N.S.) **24** (2018), no. 3, 2351–2454, arXiv:1608.03227.
4. (with J. Rasmussen, V. Shende; appendix by E. Gorsky) *The Hilbert scheme of a plane curve singularity and the HOMFLY homology of its link*, Geom. Topol. **22** (2018), no. 2, 645–691, arXiv:1201.2115.
3. (with E. Gorsky, J. Rasmussen, V. Shende) *Torus knots and the rational DAHA*, Duke Math. J. **163** (2014), no. 14, 2709–2794, arXiv:1207.4523.
2. (with D. Maulik, A. Okounkov, R. Pandharipande) *Gromov–Witten/Donaldson–Thomas correspondence for toric 3-folds*, Invent. Math. **186** (2011), no. 2, 435–479, arXiv:0809.3976.
1. (with D. Maulik) *Quantum cohomology of the Hilbert scheme of points on A_n-resolutions*, J. Amer. Math. Soc. **22** (2009), 1055–1091, arXiv:0802.2737.

The complete list of 56 papers is available at oblomkov-math.github.io.

EDITORIAL SERVICE

2025–	**Managing Editor, *Galileo***
	Mathematical Sciences Publishers (MSP); first issue 2027.

Referee for Annals of Mathematics, Inventiones, Journal of the AMS, Communications in Mathematical Physics, Advances in Mathematics, Compositio, Selecta, IMRN, and Transactions of the AMS. Grant proposal review for NSERC, NSA, Swiss NSF, NWO, and the Newton Institute; three NSF panels.

SELECTED CONFERENCE ORGANIZATION

2025	Month-long program, Bernoulli Center, EPFL, Lausanne.
2023	Two-month program, Simons Center for Geometry and Physics.
2021–2023	AIM research community and AIM workshop.
2019–2022	FRG workshops at UC Davis, UMass Amherst and the University of Oregon.
2014–2018	Workshops at Banff, Amherst and AIM.

PH.D. ADVISING

2024	Arthur Wang.
2016	Thomas Shelly — Visiting Assistant Professor, Mount Holyoke College.
2015	Tobias Wilson — Bloomberg.
Current	Pranav Kalkunte; Chuijiao Zhang.

Mentored UMass REU teams in the summers of 2013, 2015–2019 and 2025.